

about 6mm. Divyansh wondered why because his movement was about 10mm despite the fact that he was much more experienced than me." Lower the barrel movement, more the time for the shooter to hit the bullseye.

This observation made Jindal think about whether she had an advantage over her senior male counterpart. It turned out she did, and she wasn't the only one.

According to coaches and experts, rifle shooting rewards isometric strength — which is the ability to overcome resistance — rather than the dynamic strength seen in most other sports.

Interestingly, the physical anatomy of women provides female shooters a physical advantage, contrasting with almost every other sport where men have the edge.

Over the years, women have consistently outscored men in the 10m air rifle event. The Paris 2024 Olympics is no different. The qualification cut-off (eighth place score) in the men's 10m air rifle event at Chateauroux was 629.8. The same score would have placed only 12th in the women's event, where the cut-off was 631.3.

It was the same scenario at the three ISSF World Cups — the highest standard of competition in shooting — this year. The qualification cut-off for women at the Cairo World Cup was 632.3, compared to 630 for men. At the Baku World Cup, it was 631.2 for women and 630.5 for men, while at the Munich World Cup, it was 632.5 for women and 631.5 for men.

This pattern was also noticed in the mixed team event in Paris. In qualification, only Kazakhstan and Mexico finished in the top 10 with the man outscoring the woman in the pair. Kazakhstan's male competitor, Islam Satpayev, had the highest score, but the next five highest scores belonged to women. In contrast, the six worst scores were all posted by men.

This isn't a new trend. At the Tokyo Olympics, researchers discovered no statistical difference between men and women in air rifle competitions. In contrast, men did slightly better in

air pistol, trap, and skeet competitions.

Even a competition with no statistical difference in performance between men and women is remarkable. In nearly every other sport (with the exception of equestrian, where men and women compete in the same competition), men generally have the advantage.

The physical advantages that men enjoy in most sports, such as taller frames and more muscular upper bodies, prove to be a disadvantage in rifle shooting.

For instance, the men's world record in the 100m is nearly nine per cent faster than the women's over the same distance.

"Normally in sports, men have between 5-12 per cent more performance than women in the same category. This isn't happening in rifle shooting," says Professor Daniel Mon Lopez of the University of Madrid, who studied the phenomenon in his publication — *Recent Changes in Women's Olympic Shooting and Effects in Performance* — released just before the Tokyo Olympics.

Daniel Burger, the chief coach of the Swiss shooting team, which won a bronze medal in the women's 10m



Unique advantage: The more seasoned Divyansh Panwar (above) realised that the rifle barrel movements on the SCATT reading for Ramita Jindal (left) was significantly lesser than his own, despite her lack of experience. Turns out there was more to it than meets the eye. PTI

air rifle event through Audrey Gogniat, has no doubt why that is so.

"If there is a sport that is almost designed for a woman, it would have to be the 10m air rifle competition," he says. "It is a precision sport — it isn't your physical strength that is most important here. In fact, the three most important factors for shooters in the air rifle event are balance, fine motor skills, and mental control, and I think women have the edge here," he says.

This is mostly true, he says, when it comes to natural anatomical advantages. "Generally, women are shorter than men, have narrower shoulders, and wider hips. This is great for a shooter. A smaller shooter is more stable. And if your weight is distributed lower in your body — as is generally the case for women — you have a lower centre of gravity, which allows you to stay balanced.